

Data Visualization Dashboard Design Process

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SDG 6: Safe Water and Sanitation Dashboard - Design Process

Target Audience

Policymakers, environmental analysts, and development groups that work on water and sanitation projects are the target audience for the dashboard. It also targets researchers and students who want to understand global progress on SDG 6. These audiences are better able to identify trends, gaps, and areas that require immediate attention or policy action thanks to the visual insights.

PURPOSE

The main objective of this dashboard is to analyze and present significant SDG 6 indicators related to clean water and sanitation. It aims to highlight regional differences, identify countries with severe water shortages, and promote data-driven decisions that will improve water access, sanitation, and sustainable resource management.

Data Collection

The dataset was sourced from the United Nations Sustainable Development Goals (SDG) Global Database, which is mainly concerned with Goal 6: Clean Water and Sanitation. It supplies figures to assess world development in making water and sanitation available and managing them sustainably. The dataset spans from 2000 to 2024, thus illustrating not only the current state of water and sanitation availability in different regions and countries, but also the trends and patterns regarding access to these vital resources through the years.

The analysis uses five key indicators:

- 6.1.1: Proportion of population using safely managed drinking water services (%)
- 6.2.1: Proportion of population using safely managed sanitation services (%)
- 6.3.1: Proportion of safely treated domestic wastewater flows (%)
- 6.4.2: Level of water stress: freshwater withdrawal as a proportion of available freshwater resources (%)
- 6.5.1: Degree of integrated water resources management implementation (%)

These indicators are important because they combine to quantify the different aspects of water access, sanitation behavior, wastewater management, and sustainable water resource use—the three main dimensions for good evaluation of SDG 6 progress. Nevertheless, the dataset is not without its limitations and gaps.

Data Preparation and cleaning

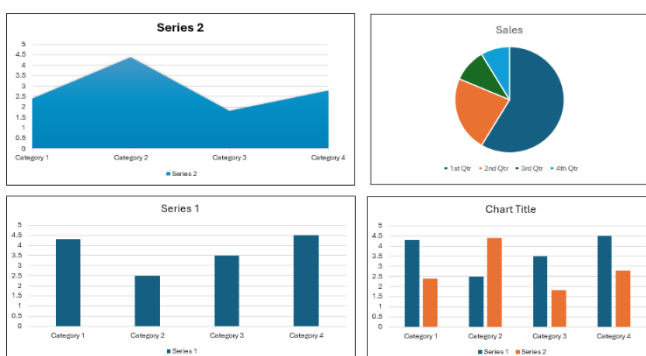
prior to visualization preprocessing involved taking away duplicates, restricting to the 2018–2024 period, and dealing with missing values. These processes made it possible for Microsoft Power BI to do a more accurate and consistent analysis

- unwanted columns in the dataset were removed to maintain clarity and to help us keep the focus on the most relevant variables only.
- The value column was then sorted to show the values from lowest to highest, to identify the null values in the column. After identifying them all null values of the column was removed.
- SeriesDescription column together with GeoAreaName and TimePeriod columns were selected together to act as a unique value(to act as the primary key) and the duplicates were removed to ensure that there only one value for a country from each year under each series discretion.
- New measurements such as, managementscore, wastewater, waterstress, wateraccess and sanitation were added to create the comparison charts.
- Even though the data set was not altered only years from 2018 – 2024 were used in the charts to make sure the charts include the latest data.

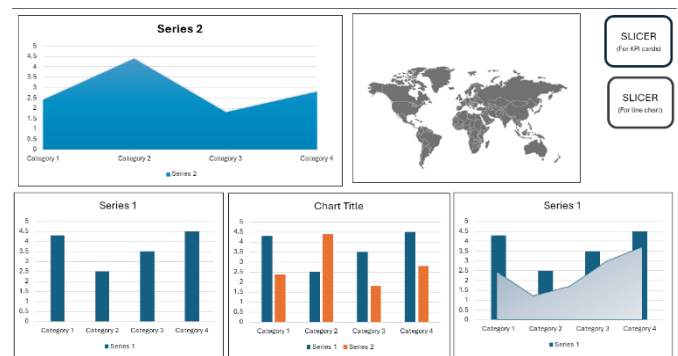
Sketching & prototyping

prior to building the dashboard the group members sketched the basic layout with some chart ideas.

- Creating the single-screen structure and outlining the primary chart regions was the main goals of the first sketch. It contained two bar charts, a pie chart, and a line chart for trends. The group's first criticism was that the sketch was too simple. Without any interactive features or a clear visual overview of current performance, it was solely descriptive and would not satisfy the audience's need useful insights.
- In order to fix the lack of interaction, the second sketch focused on incorporating the control elements of the dashboard. Two slicers were placed to the top banner to verify their position and visibility. The slicers were not yet assigned to any charts, this was a strategic step to complete the controls' visual structure to create a more interactive dashboard.



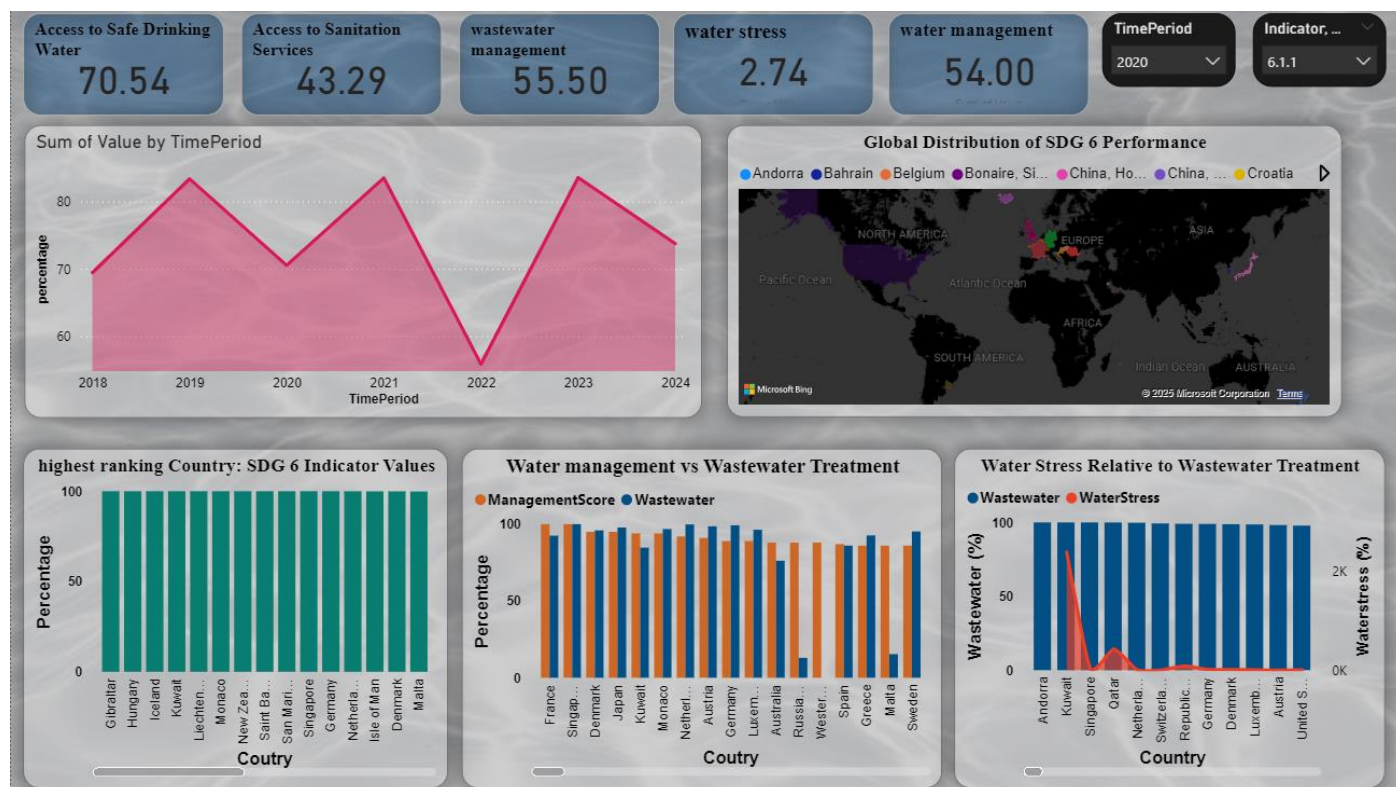
Sketch 01



Sketch 02

the First Digital Prototype was created right after the initial sketching stage, In order to evaluate the conceptual layout and the initial chart selections against actual data. The team created the initial prototype using the structure from the second concept with the addition of DATA CARDSs on the top of the dashboard.

Prototype of the dashboard



A major critique was the data cards lack country context and the unsorted ranking bar chart, which failed to clearly communicate the "best" performer, necessitating further refinement.

Iteration & Refinement

Multi-row cards replaced the data cards to show the highest performing country/countries for each indicator. Orange color background was selected for the water stress multi-row card to show that water stress being high is not good for a country.

- The “highest ranking country” bar chart was replaced with a doughnut chart which shows the lowest performing countries of the indicator 6.2.1, which allows the dashboard to better represent the SDG 6.
- A main heading was added to the dashboard and the chart headings were updated to further clarify the content.
- Colores and the background was changed improve the clarity and the esthetics of the dashboard.
- These refinements considerably enhanced the dashboard's aesthetic appeal, informative value, and ensured a strong connection with the SDG 6 objectives.

Final dashboard design.

